VIT Vidyalankar Institute of Technology
Accredited A+ by NAAC

Version 09/27-09

Consolidated Academic Administration Plan for the Course

Natural Language Processing (elective as per NBA) Sem. VII – Program Information Technology 2026-27 Odd Semester

Faculty - Dr. Suja Jayachandran (Cluster Mentor) & Rasika Ransing

The academic resources available in VIT –

VMIS (ERP)	V-Refer and V-Live	VIT Library	VAC & MOOC Courses
Institute & Department Vision and Mission	Former IA question papers and solutions (prepared by faculty)	Former IA question papers solutions - hardcopy	Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs
Program Educational Objectives (PEO)	MU end semester examination question papers and solutions (prepared by faculty)	MU end semester exam question paper & solutions - by faculty, hardcopy	
Program Specific Outcome (PSO)	Class notes and Digital Content for the subject (scanned / typed by faculty)	All text books, reference books, e -books mentioned in the syllabus & AAP	Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified
Program Outcome (PO)	Comprehensive question bank, EQ, GQ, PPT, Class Test papers	Technical journals and magazines for reference	
Departmental Knowledge Map	Academic Administration Plan & Beyond Syllabus Activity report	VIT library has many resources e:g :- IEEE, Nimbus, xplore, EBSCO etc.	Watch former lectures captured in LMS at VIT

1.a Course Objectives (Write in detail – as per NBA guidelines)

Cognitive	What do you want students to know?	Students should know the basic concepts of Natural Language Processing and algorithmic description of main language levels such as morphology, syntax, semantics, and pragmatics.
Affective	What do you want students to think / care about?	Implement various language Models.
Behavioural	What do you want students to be able to do?	Design and implement applications based on natural language processing.

Advice to Students:

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if, what you understand is correct.

The v-Refer Link

https://vrefer.vidyalankar.edu.in/index.php/apps/files/files/921184?dir=/vit%20vrefer/INFT/SEM%20VII/2026-27/verticals/PC_PEC/subjects/NLP%20%28PEIT41T%26P%29/RSR

Creation of microsite or teams link – (Division wise)

Batch-1 -

https://teams.cloud.microsoft/l/team/19%3ANV3ocSmPIjPZO94QYXuwP7ltn3NKnvBVTtiTsn0RuTs1%40thread.tacv2/conversations?groupId=72df016b-ca1a-4d0d-8e3a-f9879c8b8b13&tenantId=c7b00d7f-ad99-442a-b12f-c2c912044fdc

Batch-2

https://teams.cloud.microsoft/l/team/19%3AFT9B-2uzaPAZa-uaqVN95sCUD8P2n3vxdl_fLBlH-Js1%40thread.tacv2/conversations?groupId=4b4d3084-c814-44ca-9a40-3e58dd5b9d63&tenantId=c7b00d7f-ad99-442a-b12f-c2c912044fdc

Collaboration Policy:

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the internet, you may use their idea for the solution; provided you acknowledge the source (name and page in the book or the website, if the idea is found on the internet). Even though you are allowed to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible, please ask the teacher.

1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

CO No.	Statements	Related Module/s
CO1	Recall fundamental concepts of NLP such as stages of language processing, grammar, and basic text preprocessing techniques like tokenization and stemming.	1, 2, 3, 4, 5, 6
CO2	Explain morphological, syntactic, and semantic components of language using appropriate models and tools.	1, 2, 3, 4, 5, 6
CO3	Apply parsing algorithms and semantic techniques (like WSD and discourse resolution) to analyze linguistic structures.	2, 3, 4, 5
CO4	Analyze the structure and relationships in lexical resources (e.g., WordNet, BabelNet) and evaluate semantic similarity and ambiguity.	3
CO5	Evaluate the performance of various neural language models and word embedding techniques on language tasks.	1, 2, 4, 5, 6
CO6	Design and implement real-world NLP applications like machine translation or sentiment analysis using advanced deep learning models.	6

1.c Course Outcome (CO) Statements mapping with Learning Outcome (LO)

	Mapped to Learning Outcomes
CO1	1.1, 1.2, 2.1, 3.1, 4.1, 5.1, 6.1
CO2	1.3, 2.2, 3.2, 4.2, 5.2, 6.2
CO3	2.3, 3.3, 4.3, 5.3
CO4	3.4
CO5	1.4, 2.4, 4.4, 5.4, 6.3
CO6	6.4

1.d Mapping of COs with POs (As per new detailed Syllabus)

(List of POs is available in V-refer)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	Knowledge A	Analysis	Design	Investigation	Modern Tools	Society	Environment &sustainability	Ethics T	Teamwork C	Communication	Project Mgt	Lifelong learning
CO 1	7/53	2/53	-	-	-	1/53	-	-	-	-	-	4/53
CO 2	5/53	7/53	1/53	1/53	-	-	-	-	-	-	-	-
CO 3	-	2/53	3/53	1/53	3/53	-	-	-	-	-	-	-
CO 4	-	1/53	1/53	1/53	-	-	-	-	-	-	-	-
CO 5	-	2/53	3/53	5/53	-	-	-	-	-	-	-	-
CO6	-	-	2/53	-	1/53	-	-	-	-	-	-	-

1.e Mapping of COs with PSOs (As per new detailed Syllabus)

	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	9/53	1/53	4/53	-
CO 2	14/53	-	-	-
CO 3	6/53	3/53	-	-
CO 4	3/53	-	-	-
CO 5	10/53	-	-	-
CO6	2/53	1/53	-	-

1.f Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course

Verticals	BSC/ESC	Program Courses	Multidisciplinary Courses	Skill Courses	HSSM	Experiential Learning	Liberal Learning
Tick suitable category		√					

Subject Code	Subject Name	**Teaching Scheme**			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/Practical	Tutorial	Total
IT41T	Natural Language Processing	2	-	-	2	-	-	2
IT41P	Natural Language Processing Lab	-	2	-	-	1	-	1

Subject Code	Subject Name				**Examination Scheme**				
		Theory			**Total (Theory)**	**Practical**			**Total (Practical)**
		ISA	MSE	ESE		ISA	ESE	ORAL	

IT41T	Natural Language Processing	15	20	40	75	-	-	-	-
IT41P	Natural Language Processing Lab	-	-	-	-	25	25	-	50

Subject Code	Subject Name	MSE-1*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PEIT41T	Natural Language Processing	Q1-5*3 =15 marks	Module 1:40%	L1,L2,L3
		Q2-10*2=20 marks	Module 2:60%	L1,L2,L3,L4

Subject Code	Subject Name	MSE-2*		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PEIT41T	Natural Language Processing	Q1-5*3 =15 marks	Module 3:50%	L1,L2,L3
		Q2-10*2=20 marks	Module 4:50%	L1,L2,L3,L4

Subject Code	Subject Name	ESE#		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PEIT41T	Natural Language Processing	Q1-10*2 =20 marks	Module 1: 10% Module 2: 15%	L1,L2,L3 L1,L2,L3
		Q2-10*2=20 marks	Module 3: 15% Module 4: 15%	L1,L2,L3,L4 L1,L2,L3
		Q3-10*2 =20 mark	Module 5: 25%	L1,L2,L3
		Q4-10*2=20 marks	Module 6: 20%	L1,L2,L3,L4

*** Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

Recommended distribution: - 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30 marks from thought provoking

1.g Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course

Divisions	Lecture (Hrs.)	Practical (Hrs.)				Tutorial (Hrs.)			
		Batch 1	Batch 2	Batch 3	Batch 4	Batch 1	Batch 2	Batch 3	Batch 4
A	2	2	2	-	-	-	-	-	-
B	2	2	2	-	-	-	-	-	-
C		2	2	-	-	-	-	-	-

1.h Office Hours (Faculty will be available in office in this duration for solving students' query)

Division	Day	Time (at least 1 Hr. / Division)	Venue (Office Room No.)
A	Tuesday	4pm-5pm	E105
B	Wednesday	4pm-5pm	E105
C	Thursday	4pm-5pm	E105

Note: Office hours should be after academic hours of students.

2.a Syllabus: Module Wise Teaching Hours and % Weightage in autonomous syllabus Question Paper

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
1	**Foundations of NLP and Language Modelling** ***Learning Objective:*** **Recall basic NLP concepts such as grammar, morphology, and N-gram language models.** **Contents:** Language, Knowledge and Grammar in language processing; Stages in NLP; Ambiguities and its types in English and Indian Regional Languages; Challenges of NLP; Applications of NLP. Basic Terms: Tokenization, Stemming, Lemmatization; Survey of English Morphology, Inflectional Morphology, Derivational Morphology; Morphological Models: Dictionary lookup, finite state morphology; Lexicon free FST Porter Stemmer algorithm; Grams and its variation: Bigram, Trigram; Simple (Unsmoothed) N-grams; N-gram Sensitivity to the Training Corpus; Unknown Words: Open versus closed vocabulary tasks; Evaluating N-grams: Perplexity; Smoothing: Laplace Smoothing, Good-Turing Discounting. ***Self-Learning Topics***: *Variety types of tools for regional languages pre-processing and other functionalities, Noisy channel models, various edit*	6	15	40	10	1.3.1, 1.4.1, 2.1.1, 2.2.3, 2.3.1, 4.3.2, 12.1.1

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
	distance, Advance Issues in Language Modelling.					
Learning Outcome-1.1	*Recall key concepts such as grammar, morphology, and stages of NLP. (1.4.1 and 12.1.1) (CO1)*					
Learning Outcome-1.2	*Explain the role of tokenization, stemming, and lemmatization in text preprocessing. (1.4.1 and 2.1.1) (CO1)*					
Learning Outcome-1.3	*Apply N-gram models to analyze and represent sequences in text. (1.3.1 and 2.3.1) (CO2)*					
Learning Outcome-1.4	*Evaluate smoothing techniques like Laplace and Good-Turing for language model performance. (4.3.2 and 2.2.3) (CO5)*					
2	**Parsing Techniques and Syntax Modelling** ***Learning Objective: Explain various parsing strategies and apply syntactic models like CYK and PCFG to analyze sentences.*** **Contents:** Parsers: Top down and bottom up; Modelling constituency; Bottom-Up Parser: CYK, PCFG (Probabilistic Context Free Grammar), Shift Reduce Parser; TopDown Parser: Early Parser, Predictive Parser ***Self-Learning Topics***: *Evaluating parsers, Parsers based language modelling, Regional languages POS tree banks*	5	20	15	15	1.4.1, 2.1.1, 2.1.2, 2.3.1 3.3.1, 3.4.2, 4.3.2
Learning Outcome-2.1	*Recall the types of parsers and their classification as top-down or bottom-up. (1.4.1 and 2.1.1) (CO1)*					
Learning Outcome-2.2	*Describe how CYK, PCFG, and Earley parsers operate for syntactic analysis. (2.1.2 and 1.4.1) (CO2)*					
Learning Outcome-2.3	*Apply parsing algorithms on given sentences to generate parse trees. (2.3.1 and 3.4.2) (CO3)*					
Learning Outcome-2.4	*Evaluate parser outputs and performance using parsing metrics. (3.3.1 and 4.3.2) (CO5)*					
3	**Semantics, and Discourse** ***Learning Objective: Analyze lexical semantics and perform***	5	20	50	15	1.4.1, 2.1.1, 2.3.1, 2.2.3, 4.3.1, 5.1.1, 12.1.1

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
	word sense disambiguation and discourse analysis. **Contents:** Introduction, meaning representation; Lexical Semantics; Corpus study; Knowledge Graphs & Ontologies; Study of Various language dictionaries like WorldNet, Babelnet; Relations among lexemes & their senses –Homonymy, Polysemy, Synonymy, Hyponymy; Semantic Ambiguity; Word Sense Disambiguation (WSD); Semantic Similarity and Relatedness. Discourse: Reference Resolution, Reference Phenomena, Syntactic & Semantic constraint on coherence; Anaphora Resolution using Hobbs Algorithm ***Self-Learning Topics***: *Dictionaries for regional languages, Topic Models, Discourse segmentation, Conference resolution*					
Learning Outcome-3.1	*Recall semantic relations like synonymy, polysemy, and hyponymy. (1.4.1 and 12.1.1) (CO1)*					
Learning Outcome-3.2	*Explain the concept of word sense disambiguation and discourse coherence. (2.1.1 and 1.4.1) (CO2)*					
Learning Outcome-3.3	*Apply anaphora resolution techniques like Hobbs' algorithm in discourse contexts. (2.3.1 and 5.1.1) (CO3)*					
Learning Outcome-3.4	*Analyze semantic ambiguity using ontologies and knowledge graphs. (4.3.1 and 2.2.3) (CO4)*					
4	**Word Representations and Embedding Models** ***Learning Objective: Implement word embedding techniques like Word2Vec, fastText, and GloVe for representing text data.*** **Content:** Word2Vec, CBOW and Skip-Gram Models, One word learning architecture, Forward pass for Word2Vec, Matrix Operations, Lec 07: Word Representation: Word2Vec & fastText, Word Representation: GloVe, Tokenization Strategies	6	15	50	15	1.4.1, 2.1.2, 2.2.3, 3.4.2, 4.3.2, 5.1.1, 12.1.1

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-4.1	*Identify various word embedding models such as Word2Vec, GloVe, and fastText. (1.4.1 and 12.1.1) (CO1)*					
Learning Outcome-4.2	*Explain CBOW and Skip-gram models with forward-pass architecture. (1.4.1 and 2.1.2) (CO2)*					
Learning Outcome-4.3	*Implement word embeddings using real-world textual data. (5.1.1 and 3.4.2) (CO3)*					
Learning Outcome-4.4	*Compare and analyze embeddings based on their contextual and subword features. (4.3.2 and 2.2.3) (CO5)*					
5	**Neural Language Models and Sequence Learning** ***Learning Objective: Evaluate deep learning models such as LSTM, GRU, and attention-based mechanisms for language modeling.*** **Content:** Neural Language Models - CNN, RNN, RNN - Based Language Model, LSTM, GRU, Sequence-to-Sequence Models, Greedy Decoding, Beam search, Other Decoding Strategies: Nucleus Sampling, Temperature Sampling, Top-k Sampling, Attention in Sequence-to-Sequence Models	5	15	0	25	1.4.1, 2.1.2, 3.3.1, 3.4.2, 4.3.1, 4.3.2, 5.1.1, 12.1.1
Learning Outcome-5.1	*Recall the architecture of neural models like RNN, LSTM, and GRU. (1.4.1 and 12.1.1) (CO1)*					
Learning Outcome-5.2	*Differentiate between greedy decoding, beam search, and sampling strategies. (2.1.2 and 4.3.1) (CO2)*					
Learning Outcome-5.3	*Apply neural models to generate or predict sequences in NLP tasks. (3.4.2 and 5.1.1) (CO3)*					
Learning Outcome-5.4	*Evaluate language models based on accuracy, fluency, and relevance in output. (4.3.2 and 3.3.1) (CO5)*					
6	**Applications of NLP in Real-World Scenarios** ***Learning Objective: Design and develop practical NLP applications like sentiment analysis and machine translation.*** **Contents:**	3	15	0	10	1.4.1, 2.2.1, 3.1.1, 3.2.2, 3.3.1, 4.3.2, 5.2.1, 6.1.1

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
	Case studies on (preferable in regional language): Machine translation; Text Summarization; Sentiment analysis; Information retrieval; Question Answering system ***Self-Learning Topics***: *Applications based on Deep Neural Network with NLP such as LSTM network, Recurrent Neural network etc.*					
Learning Outcome-6.1	*Describe common NLP applications such as MT, IR, and sentiment analysis. (1.4.1 and 6.1.1) (CO1)*					
Learning Outcome-6.2	*Explain the workflow of an NLP application using deep learning models. (3.1.1 and 2.2.1) (CO2)*					
Learning Outcome-6.3	*Assess the performance of your NLP system using relevant metrics. (3.3.1 and 4.3.2) (CO5)*					
Learning Outcome-6.4	*Develop an end-to-end NLP solution for a real-world problem. (3.2.2 and 5.2.1) (CO6)*					
	Total	**30**	**100**	**100**	**100**	

2.b Prerequisite Courses

No.	Semester	Name of the Course	Topic/s
1	V	PCIT09 (Automata Theory)	Grammar, Finite Automata
2	VI	PEIT29 (Probabilistic Graphical Model)	Hidden Markov Model

2.c Relevance to Future Courses

No.	Semester	Name of the Course
1	VIII	HMIT09T & HMIT09P (Generative AI models)

2.d

See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video)

Real Life Scenario	Concept Used
Sentiment Analysis	Tokenization, Stop word filtering, Stemming

Machine Translation	Morphological analysis, Lexical analysis
Question Answering	Question Analysis, Document retrieval, Answer extraction

3 Past Results – Division-Wise

Details	Target – DEC 2026	MAY 2025	DEC 2024	MAY 2024
Course Passing % – Average of 2 Divisions	100	98	NA	NA
Marks Obtained by Course Topper (mark/100)	72/75	70/75	NA	NA

	Division A		Division B		Division C	
Year	Initials of Teacher	% Result	Initials of Teacher	% Result	Initials of Teacher	% Result
May 2025	RSR	98	RSR	98	NA	NA
May 2024	NA	NA	NA	NA	NA	NA
May 2023	NA	NA	NA	NA	NA	NA

4 All the Learning Resources – Books and E-Resources

4.a List of Textbooks (T – Symbol for Textbooks) to be Referred by Students

Sr. No	Textbook Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Speech and Language Processing	Daniel Jurafsky, James H. Martin	Prentice Hall	2	All	Yes
2	Foundations of Statistical Natural Language Processing	Christopher D.Manning and Hinrich Schutze	MIT Press	-	3, 4	Yes

4.b List of Reference Books (R – Symbol for Reference Books) to be Referred by Students

Sr. No	Reference Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Natural Language Processing and Information Retrieval	Siddiqui and Tiwary U.S.,	Oxford University Press	2008	1	Yes
2	Multilingual natural language processing applications	Daniel M Bikel and Imed Zitouni	Pearson	2013	2, 5, 6	No
3	The Handbook of Computational Linguistics and Natural Language Processing	Alexander Clark, Chris	Wiley	2010	6	Yes

		Fox, Shalom Lappin				
4	Handbook of Natural Language Processing	Nitin Indurkhya and Fred J. Damerau	Chapman and Hall/CRC Press	2010	All	Yes
5	Natural Language Processing with Python	Steven Bird, Ewan Klein	O'Reily	-	Lab	Yes
6	A step by step tutorial: An introduction into R application and programming	Niel J le Roux, Sugnet Lubbe	-	-	Lab	Yes

4.c List of E - Books (E – Symbol for E-Books) to be Referred by Students

Sr. No	E- Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Natural Language Processing Succinctly	Joseph D. Booth	Syncfusion	-	2, 3	Yes

4.d Reading latest / top rated research papers (at least 5 papers)

Name of Paper	Name of Authors (Background)	Published in		Problem Statement	Available in our Library
		Date	Journal		
Leveraging large language models for word sense disambiguation	Jung H. Yae, Nolan C. Skelly, Neil C. Ranly & Phillip M. LaCasse	2025	Neural Computing and Applications	Evaluate the WSD capabilities of generative LLMs using three task-specific techniques, assessing their effectiveness in both known and novel word contexts, and examining the implications for real-world NLP applications.	Available on Teams
Attention Is All You Need	Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser	2017	31st Conference on Neural Information Processing Systems (NIPS 2017)	Develop and implement a Transformer model utilizing scaled dot-product attention, multi-head attention, and positional encoding, and compare its performance with LSTM and GRU models on a benchmark NLP dataset. Evaluate the models in terms of classification/translation accuracy, training time, inference speed, computational complexity, and ability to capture long-range contextual dependencies. The objective is to demonstrate the advantages of attention-only architectures over traditional sequential neural networks.	Available on Teams
NYCU-NLP at SemEval-2025 Task 11: Assembling Small Language Models for Multilabel Emotion	Zhe-Yu Xu, Yu-Hsin Wu and Lung-Hao Lee	2025	Proceedings of the The 19th International Workshop on Semantic Evaluation (SemEval-2025)	Emotion recognition from text remains a challenging Natural Language Processing task because a single sentence can simultaneously express multiple emotions with varying levels of intensity. There is a need for an efficient yet accurate framework that leverages instruction-tuned Small Language Models (SLMs) and	Available on Teams

Detection and Intensity Prediction				ensemble learning techniques to perform multilabel emotion detection and emotion intensity prediction while maintaining competitive performance with significantly lower computational requirements. This research aims to design, fine-tune, and evaluate an ensemble of SLMs for multilingual emotion analysis.	
LLM-based conversational agents for behaviour change support: A randomised controlled trial examining efficacy, safety, and the role of user behaviour	Selina Meyer, David Elsweiler	2025	International Journal of Human - Computer Studies	Although Large Language Models (LLMs) have shown promise in conversational support for behaviour change, existing chatbots often lack structured counselling strategies, personalization, and safeguards against harmful or misleading responses. There is a need for an LLM-based conversational agent that applies Motivational Interviewing (MI) principles to encourage self-reflection while ensuring safe, empathetic, and effective conversations. Such a system should also analyze how different user interaction behaviours influence the quality and success of behaviour change support.	Available on Teams
Automated Research Review Support Using Machine Learning, Large Language Models, and Natural Language Processing	Vishnu S. Pendyala, Karnavee Kamdar, Kapil Mulchandani	2025	Electronics Special Issue Data-Centric Artificial Intelligence: New Methods for Data Processing	Existing review workflows rely heavily on manual reviewer assignment and subjective evaluation, which can introduce inefficiencies, inconsistencies, and bias. There is a need for an intelligent system that leverages Machine Learning, Large Language Models (LLMs), and Natural Language Processing (NLP) to automate reviewer recommendation, predict paper quality and acceptance, and assist editorial decision-making. Such a system can improve the speed, consistency, and scalability of the peer-review process while supporting human reviewers rather than replacing them.	Available on Teams
Benchmarking large language models for biomedical natural language processing applications and recommendations	Qingyu Chen, Yan Hu, Xueqing Peng, Qianqian Xie, Qiao Jin, Aidan Gilson, Maxwell B. Singer, Xuguang Ai, Po-Ting Lai2, Zhizheng Wang, Vipina K. Keloth, Kalpana Raja, Jimin Huang, Huan He,	2025	Nature communications	The increasing volume of biomedical literature has made manual extraction, analysis, and interpretation of medical knowledge slow and resource-intensive. Although Large Language Models (LLMs) have shown strong capabilities in general NLP tasks, their reliability and effectiveness in biomedical NLP applications remain uncertain due to limited domain-specific evaluation and issues such as hallucinations, inconsistencies, and missing information. There is a need for a comprehensive framework to benchmark LLMs across diverse biomedical NLP tasks and compare them with traditional fine-tuned biomedical models. Such an evaluation	Available on Teams

	Fongci Lin, Jingcheng Du, Rui Zhang 5,6, W. Jim Zheng, Ron A. Adelman, Zhiyong Lu, Hua Xu			can identify the strengths, limitations, and practical recommendations for deploying LLMs in real-world biomedical applications.	

4.e **Based on research paper an identify the current Problem statement**

Problem Statement			Used in			
	Quiz	Assignment	Mini Project	Poster Presentation	Test	Any Other
Leveraging large language models for word sense disambiguation - Evaluate the WSD capabilities of generative LLMs using three task-specific techniques, assessing their effectiveness in both known and novel word contexts, and examining the implications for real-world NLP applications.						Experiment
Attention Is All You Need – Develop and implement a Transformer model using scaled dot-product attention, multi-head attention, and positional encoding, and compare its performance with LSTM and GRU models on a benchmark NLP dataset. Evaluate the models based on accuracy, training time, inference speed, computational complexity, and their ability to capture long-range contextual dependencies to demonstrate the effectiveness of attention-based architectures over traditional sequential models.		√				
NYCU-NLP at SemEval-2025 Task 11: Assembling Small Language Models for Multilabel Emotion Detection and Intensity Prediction – Emotion recognition from text is challenging because a single sentence can express multiple emotions with varying intensity levels, making accurate prediction difficult. This project aims to develop and evaluate an ensemble of instruction-tuned Small Language Models (SLMs) for multilingual multilabel emotion			√			

detection and emotion intensity prediction, achieving competitive performance with lower computational cost.						

4.f **Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements**

Name of the Company	To be / Contacted for		
	Student Internship	Student Final Placement	Faculty Internship
Hanooman AI	√	√	

4.g **Identify suitable relevant TOP Guest Speakers from Industry, Example: - (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)**

Name of the Identified Guest Speaker	Designation	Name of the Company
Prof. Pawan Goyal	Associate Professor, Indian Institute of Technology, Kharagpur,	NPTEL/Swayam

4.h **Identify relevant technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]**

Name of the Relevant Technical Competition Identified to participate	Organized by	Date of the Event
Smart India Hackathon (SIH) 2026	Ministry of Education, Government of India	Yet to be declared

4.i **Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication**

University	Name of the Course	Name of Faculty	Type of Collaboration		
			Exchange of Lecture Material	Joint Publication/ Research	Other
IIT Kharagpur	Natural Language Processing	Prof. Pawan Goyal	https://archive.nptel.ac.in/noc/courses/noc19/SEM2/noc19-cs56/	-	-

4.j **Module Best Available in – Title of the best resource [from *4.a* to *4.d* in this AAP] and other details as necessary**

Module No.	Title of the Module	Textbook	Mention the Title					
			Reference Book	E-books	Journal	E-Journal	Available in our Library	V-refer
1	Foundations of NLP and Language Modelling	1, 2	1, 2	1	Natural Language		Yes	Yes

					Processing Journal			
2	Parsing Techniques and Syntax Modelling	1, 2	1, 2, 3	1	Natural Language Processing Journal		Yes	Yes
3	Semantics, and Discourse	1, 2	1, 2, 4	1	Natural Language Processing Journal		Yes	Yes
4	Word Representations and Embedding Models	1, 2	1, 2	1	Natural Language Processing Journal		Yes	Yes
5	Neural Language Models and Sequence Learning	1, 2	1, 2	1	Natural Language Processing Journal		Yes	Yes
6	Applications of NLP in Real-World Scenarios	1, 2	1, 2,5	1	Natural Language Processing Journal		Yes	Yes

4.k **Referred to any top-rated university in that subject for content**

University	Name of the Course	Name of Faculty	Date of Delivery of the Course	Remarks
Stanford University	CS224n: Natural Language Processing with Deep Learning	Christopher D Manning	Winter 2025	https://web.stanford.edu/class/cs224n/

4.l **Faculty received any certification related to this subject. List of Certifications Identified / Done**

Course	Certifying Agency	No. of Hours	Level of the Course		Certification		Remarks
			Introductory	Advance Skill Development	Done on	Proposed to be on	
Natural Language Processing	SWAYAM NPTEL	12 weeks	√		2019		
Introduction to Large Language Models (LLMs)	SWAYAM NPTEL	12 weeks		√	2025		

4.m Completed subject wise/cluster wise training with cluster mentor. List of relevant Refresher Course Identified / Done

Course	Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/University other than MU)	Certification		Remarks
		Done on	Proposed to be on	
Pedagogy	Instructional Planning and Delivery (NITTR)	September 2021		
PBL				
Sub. Content Training	Introduction to Large Language Models (LLMs)	2025		

4.n Best Practices Identified and adopted

No.	Item	Best Practices Identified		
		IIT Bombay	MIT,USA	Washington.edu
1	Microsite	https://www.cse.iitb.ac.in/~cs626-449/	https://www.mit.edu/~jda/teaching/6.864/	http://courses.washington.edu/ling567/
2	Video Lectures	√		
3	Assignments	√		
4	Mini Project		√	
5	Assessment Metric	√		
6	Quizzes			√
7	Labs/ Practical (PBL)		√	
8	Tests	√		
9	Peer Assessment		√	
10	Any Other			

4.o Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

No.	Websites / Links	Module Nos.
1	https://www.kaggle.com/c/word2vec-nlp-tutorial/	6
2	https://www.cse.iitb.ac.in/~cs626-449/	All
3	https://onlinecourses.nptel.ac.in/noc20_cs87/preview	4,5

4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

Sr. No.	MOOC Course Link	Course conducted by – Person / University / Institute / Industry	Course Duration	Certificate (Y / N)

1	https://www.kaggle.com/learn/natural-language-processing	Kaggle	Self-paced	Y
	https://www.coursera.org/specializations/natural-language-processing	Deeplearning.ai	Self-paced	Y
	https://www.coursera.org/projects/deep-learning-nlp-gpt-2	Coursera	Self-paced	Y

5 Consolidated Course Lesson Plan

	From (date/month/year)	From (date/month/year)	Total Number of Weeks
Semester Duration	06/07/2026	23/10/2026	16

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
1	1	1	Language, Knowledge and Grammar in language processing; Stages in NLP; Ambiguities and its types in English and Indian Regional Languages; Challenges of NLP; Applications of NLP. Basic Terms: Tokenization, Stemming, Lemmatization; Survey of English Morphology, Inflectional Morphology, Derivational Morphology;		CO1	L1	NA	T1/Chp 3/Pg.- 85-104
2	2	1	Morphological Models: Dictionary lookup, finite state morphology; Lexicon free FST Porter Stemmer algorithm; Grams and its variation: Bigram, Trigram; Simple (Unsmoothed) N-grams; N-gram Sensitivity to the Training Corpus; Unknown Words: Open versus closed vocabulary tasks; Evaluating N-grams: Perplexity; Smoothing:		CO1	L2	NA	T1/Chp 2/Pg.- 246

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
			Laplace Smoothing, Good-Turing Discounting.					
3	3	2	Parsing Techniques and Syntax Modelling. Parsers: Top down and bottom up; Modelling constituency;		CO2	L3	NA	T1/Chp 8/Pg.-315-334, 338-341
4	4	3	Bottom-Up Parser: CYK, PCFG, (Probabilistic Context Free Grammar)		CO2	L3	NA	T1/Chp 8/Pg.-315-334, 338-341
5	5	3	Shift Reduce Parser; Top Down Parser: Early Parser, Predictive Parser		CO2	L3		T2/Chp10 ,16/Pg.-356, 589-596
6	6	3	Semantics, and Discourse Introduction, meaning representation; Lexical Semantics; Corpus study; Knowledge Graphs & Ontologies; Study of Various language dictionaries like WorldNet, Babelnet; Relations among lexemes & their senses – Homonymy, Polysemy, Synonymy, Hyponymy; Semantic Ambiguity;		CO3	L4	NA	T1/Chp 15,16,17/ Pg.-571-594,618-630,662-671
7	7	3	Word Sense Disambiguation (WSD); Semantic Similarity and Relatedness.		CO3	L4	NA	T1/Chp 15,16,17/ Pg.-571-594,618-

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
								630,662-671
8	8	3	Discourse: Reference Resolution, Reference Phenomena, Syntactic & Semantic constraint on coherence; Anaphora Resolution using Hobbs Algorithm		CO3	L4	NA	T1/Chp18/Pg.-697-706
9	9	4	Word Representations and Embedding Models Word2Vec, CBOW and Skip-Gram Models, One word learning architecture, Forward pass for Word2Vec, Matrix Operations,		CO4	L3	NA	https://onlinecourses.nptel.ac.in/noc20_cs87/preview
10	Ganpati Vacation							
11	10	4	Word Representation: Word2Vec & fastText, Word Representation: GloVe, Tokenization Strategies		CO4	L3	NA	https://onlinecourses.nptel.ac.in/noc20_cs87/preview
12	11	5	Neural Language Models and Sequence Learning Neural Language Models - CNN, RNN, RNN - Based Language Model, LSTM, GRU, Sequence-to-Sequence Models, Greedy Decoding, Beam search,		CO5	L3	NA	https://onlinecourses.nptel.ac.in/noc20_cs87/preview

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Mapping Bloom Taxonomy level	Recommended Prior Viewing / Reading	
							Lecture No. (on LMS)	Chapter No./ Books/ Web Site
13	12	5	Other Decoding Strategies: Nucleus Sampling, Temperature Sampling, Top-k Sampling, Attention in Sequence-to-Sequence Models		CO5	L3	NA	https://onlinecourses.nptel.ac.in/noc20_cs87/preview
14	13	6	Applications of NLP in Real-World Scenarios Case studies on (preferable in regional language): Machine translation; Text Summarization; Sentiment analysis;		CO6	L5	NA	T1/Chp 21
15	14	6	Information retrieval; Question Answering system		CO6	L5	NA	R1/Chp 9 -11
16	15		Revision					

6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

Theory (ISA= 15)												Total
Course	Assignment -1	Assignment -2	Assignment -3	Assignment -4	Assignment -5	Assignment -6	Assignment -7	Assignment -8	Assignment -9	Assignment -10	Assignment -11	
5	1	1	1	1	1	1	1	1	1	1	1	15

Practical (ISA=25)												Total
Mini Project	Practical -1	Practical -2	Practical -3	Practical -4	Practical -5	Practical -6	Practical -7	Practical -8	Practical -9	Practical -10	Practical -11	
15	1	1	1	1	1	1	1	1	1	1	1	25

Class Participation	MSE-1	MSE-2	ESE	Total
15	20	20	40	75

7 Assignments / Tutorials Details

Assignment/ Tutorial No.	Title of the Assignments / Tutorials	CO Map	Mapping Bloom Taxonomy Level	Assignment/ Tutorials given to Students on	Assignments to be submitted back on
0	Assignment -0	-	-	08-07-2026	15-07-2026
1	Assignment 1 – Poster Presentation	1	L1, L2, L4	15-07-2026	22-07-2026
2	Assignment 2 – Evolution & Comparison of LLMs	1	L1, L2, L3	22-07-2026	29-07-2026
3	Assignment 3 – Language Models, FSA & Case Studies	2	L1, L2, L3	29-07-2026	05-08-2026
4	Assignment 4 – Python Libraries & Bigram Problems	2	L1, L3, L4	05-08-2026	12-08-2026
5	Assignment 5 – HMM POS Tagging & Parsing	3	L1, L2, L3	12-08-2026	19-08-2026
6	Assignment 6 – Word Sense Disambiguation & Lexical Resources	3	L1, L2, L3	19-08-2026	26-08-2026
7	Assignment 7 - WSD Algorithms Review	4	L1, L2, L3	26-08-2026	02-09-2026
8	Assignment 8 – Online Certification	4	L1, L2, L3	02-09-2026	09-09-2026
9	Assignment 9 – Word Embeddings Case Study	5	L1, L2, L3	09-09-2026	16-09-2026

10	Assignment 10 – Neural Language Models Case Study	5	L1, L2, L3	23-09-2026	30-09-2026
11	Assignment 11 – Research Paper Summary	6	L1, L2, L3,L5	30-09-2026	07-10-2026
12	Assignment 12 - Seminar Presentation	6	L2, L3	07-10-2026	15-10-2026

Analysis of Assignment / Tutorial Questions and Related Resources

Assignment / Tutorial No.	Week No.	Type* (√)			Module No.	Based on #			Question Type (√)	
		OT	CS	DTP		Textbook	Reference Book	Other Learning Resource	Real Life Assignments	Thought Provoking
1	2		√		1	1			√	
2	3	√			1	1			√	
3	4		√		2		1			√
4	5	√			2		1		√	
5	6	√			3		1			√
6	7			√	3	1			√	
7	8			√	4	1				√
8	9			√	4	1				√
9	10		√		5		1			√
10	11	√			5		1		√	
11	12			√	6			1	√	

* Tick (√) the Type of the Assignment: Online Tools (OT); Collaborative Assignments (CS); Design /Thought provoking (DTP)

Write number for textbook, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

Tests	Test Dates	Module No.	CO Map	MSE Question Paper Pattern	Policy
ISE	5th week	1,2	1,2	Descriptive/Numerical Question:5*3=15marks	
Pop Quiz	7th week	1,2,3	1,2,3	Multiple Choice Question	
Open Book Test	9th week	4	4	Case study based/numerical question:10*2=20 marks	

Take Home Test	11th week	5	5	Descriptive/Numerical Question:5*3=15marks	

9. Practical Activities

Practical No.	Module No.	Title of the **Experiments**	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	1	Develop a Python program to perform text pre-processing on English and Marathi/Hindi text by implementing techniques such as text cleaning, lowercasing, tokenization, stop word removal, punctuation and number removal, stemming, and lemmatization. ***Learning Outcomes:*** *A learner will be able to:* *1.2 Explain the role of tokenization, stemming, and lemmatization in text preprocessing. (1.4.1) (CO1)*			Text Preprocessing	CO1
2	2	Feature Extraction: TF-IDF, N-grams, and Word Embeddings ***Learning Outcomes:*** *A learner will be able to:* *1.3 Apply N-gram models to analyze and represent sequences in text. (1.4.1) (CO2)* *4.3 Implement word embeddings using real-world textual data. (4.3.1) (CO3)*			Feature Extraction	CO3
3	2	Named Entity Recognition using Deep Learning techniques/ Transformer - based techniques/ LLMs ***Learning Outcomes:*** *A learner will be able to:* *1.1 Recall key concepts such as grammar, morphology, and stages of NLP. (1.4.1) (CO1)* *2.3. Apply parsing algorithms on given sentences to generate parse trees. (2.3.1) (CO3)*			Syntax analysis	CO3
4	3	Develop a Word Sense Disambiguation Framework using Large Language Models for Low-Resource Indian Languages. ***Learning Outcomes:*** *A learner will be able to:* *3.2 Explain the concept of word sense disambiguation and discourse coherence. (1.4.1) (CO2)* *3.4 Analyze semantic ambiguity using ontologies and knowledge graphs. (3.4.1) (CO4)*			Semantic analysis	CO4

5	4	Develop a Retrieval-Augmented Generation (RAG) based Question Answering System. ***Learning Outcomes:*** *A learner will be able to:* *4.3 Implement word embeddings using real-world textual data. (5.1.1 and 3.4.2) (CO3)*		Y	Semantic Retrieval	CO3
6	5	Comparative Performance Analysis of RNN, LSTM, GRU and Transformer Models for Text Classification – News Classification ***Learning Outcomes:*** *A learner will be able to:* *6.2 Explain the workflow of an NLP application using deep learning models. (3.1.1 and 2.2.1) (CO2)* *6.3 Assess the performance of your NLP system using relevant metrics. (3.3.1 and 4.3.2) (CO5)* *6.4 Develop an end-to-end NLP solution for a real-world problem. (3.2.2 and 5.2.1) (CO6)*		Y	Neural networks	CO5
7	6	Design a sentiment analysis system for product reviews in a regional language. ***Learning Outcomes:*** *A learner will be able to:* *6.1 Describe common NLP applications such as MT, IR, and sentiment analysis. (1.4.1) (CO1)* *6.2 Explain the workflow of an NLP application using deep learning models. (2.2.3) (CO2)* *6.4 Develop an end-to-end NLP solution for a real-world problem. (3.4.2) (CO6)*			NLP Pipeline	CO6
8	All	Mini Project			End to end project	CO6

10 Uncovering syllabus with different Activities.

No.	Type of the Activity	Activities	Number of beneficiaries	Other Details – guest profile, feedback, mark sheet, report
1	**Experiential learning/Interaction with Outside World**	1- Guest Lectures by Industry Expert	All students	Mr. Laxmikant Tiwari, AVP, IGP.com
		2- Workshops		
		3- Mini Project	All students	
		4- Industrial Visit		

		5- Any other activity		
2	**Collaborative & Group Activity**	6- Poster Presentation	All students	Group activity
		7- Minute Papers	All students	
		8- Students Seminars		
		9- Students Debates		
		10- Panel Discussion / Mock GD		
		11- Mock Interview		
		12- Any other activity		
3	**Co-Curricular Activity**	13- Informative videos (NPTEL/YouTube /TEDx/ MIT OW/edX)	All students	https://archive.nptel.ac.in/courses/106/105/106105158/ Lecture 39 and lecture 40
		14- Lecture Capture Usage		
		15- Any other activity		
4	**Tests & Assessments**	16- Class Tests/ Weekly Tests		
		17- Pop Quiz		
		18- Mobile App Based Quiz		
		19- Open Book Test	All students	Numerical based Question
		20- Take Home Test		

		21- Any other activity		

11 AAP

No.	Programme	Course	Uploaded on V-refer	Date
1	Information Technology	Natural Language Processing		

12 Lecture Guide

No.	Programme	Course	Uploaded on V-refer	Date
1	Information Technology	Natural Language Processing		

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

Rasika Ransing		

Mr. Laxmikant Tiwari External Industry Mentor	Dr. Bharti Joshi External Academic Mentor	Dr. Suja Jayachandran VIT Cluster Mentor	Dr. Vidya Chitre INFT HOD

Annexure:

Assessment Criteria of Learning Outcomes:

Learning Outcomes: The Learner will:	**Assessment Criteria:** The Learner can:	**Evaluated under ISA/MSE/ESE/LAB**
LO 1.1: *Recall key concepts such as grammar, morphology, and stages of NLP. (P.I.- 1.41 and 12.1.1) (CO1)*	• Define the terms grammar, morphology, and NLP. • List the stages of Natural Language Processing. • Identify types of morphology from examples. • Match NLP stages with their respective tasks. • Recall applications and challenges of NLP.	ISA/MSE/ESE
LO 1.2: *Explain the role of tokenization, stemming, and lemmatization in text preprocessing. (P.I.- 1.4.1 and 2.1.1) (CO1)*	• Describe how tokenization splits text into units. • Explain differences between stemming and lemmatization. • Identify preprocessing tools and techniques. • Classify tokens from sample input. • Provide examples of NLP preprocessing.	ISA/MSE/LAB
LO 1.3: *Apply N-gram models to analyze and represent sequences in text. (P.I.- 1.3.1 and 2.3.1) (CO2)*	• Construct bigram and trigram models. • Predict next word using N-grams. • Analyze sentence sequences using grams. • Implement N-gram model on a small dataset.	MSE/ESE/LAB
LO 1.4: *Evaluate smoothing techniques like Laplace and Good-Turing for language model performance. (P.I.- 4.3.2 and 2.2.3) (CO5)*	• Define smoothing and its purpose. • Differentiate Laplace and Good-Turing techniques. • Evaluate effectiveness using perplexity. • Apply smoothing to sparse N-gram data.	ESE/LAB
LO 2.1: *Recall the types of parsers and their classification as top-down or bottom-up. (P.I.- 1.4.1 and 2.1.1) (CO1)*	• Define top-down and bottom-up parsing. • List common parsers like CYK, Earley, Shift-Reduce. • Classify parsers based on parsing strategy.	ISA/MSE/ESE

	• Identify the parser type from a given scenario.	
LO 2.2: *Describe how CYK, PCFG, and Earley parsers operate for syntactic analysis. (P.I.- 2.1.2 and 1.4.1) (CO2)*	• Explain the functioning of CYK, PCFG, and Earley parsers. • Draw a parse tree using one of the techniques. • Distinguish use-cases where each parser is suitable.	MSE/ESE/LAB
LO 2.3: *Apply parsing algorithms on given sentences to generate parse trees. (P.I.- 2.3.1 and 3.4.2) (CO3)*	• Apply CYK or Earley parser to a sentence. • Create a valid parse tree. • Debug syntactic errors in input using parser output.	ESE/LAB
LO 2.4: *Evaluate parser outputs and performance using parsing metrics. (P.I.- 3.3.1 and 4.3.2) (CO5)*	• Interpret parse tree accuracy. • Assess parser output consistency. • Calculate metrics like precision and recall for parsing.	ESE
LO 3.1: *Recall semantic relations like synonymy, polysemy, and hyponymy. (P.I.- 1.4.1 and 12.1.1) (CO1)*	• Define synonymy, polysemy, and hyponymy. • Identify word relationships from a word pair list. • Match terms with their definitions using examples.	ISA/MSE
LO 3.2: *Explain the concept of word sense disambiguation and discourse coherence. (P.I.- 2.1.1 and 1.4.1) (CO2)*	• Describe what WSD is with examples. • Explain how context resolves ambiguity. • Outline discourse coherence concepts like anaphora. • Relate WSD and discourse coherence in text understanding.	ISA/MSE/ESE
LO 3.3: *Apply anaphora resolution techniques like Hobbs' algorithm in discourse contexts. (P.I.- 2.3.1 and 5.1.1) (CO3)*	• Apply Hobbs' algorithm to resolve references. • Identify antecedents in text. • Use syntactic trees for reference resolution.	MSE/ESE/LAB
LO 3.4: *Analyze semantic ambiguity using ontologies and knowledge graphs. (P.I.- 4.3.1 and 2.2.3) (CO4)*	• Interpret meaning from ontology relationships. • Identify ambiguous terms. • Resolve ambiguity using semantic web tools or graphs.	ESE/LAB
LO 4.1: *Identify various word embedding models such as Word2Vec, GloVe, and fastText. (P.I.- 1.4.1 and 12.1.1) (CO1)*	• List common word embedding models. • Match models to their architecture or output. • Distinguish contextual vs non-contextual embeddings.	ISA/MSE

LO 4.2: *Explain CBOW and Skip-gram models with forward-pass architecture. (P.I.- 1.4.1 and 2.1.2) (CO2)*	• Describe CBOW and Skip-gram mechanisms. • Explain how training happens using forward pass. • Diagram forward propagation steps.	MSE/ESE
LO 4.3: *Implement word embeddings using real-world textual data. (P.I.- 5.1.1 and 3.4.2) (CO3)*	• Use libraries to train Word2Vec or fastText. • Clean and prepare real-world corpus. • Evaluate embedding quality using similarity checks.	ESE/LAB
LO 4.4: *Compare and analyze embeddings based on their contextual and subword features. (P.I.- 4.3.2 and 2.2.3) (CO5)*	• Compare fastText and Word2Vec results. • Explain benefits of subword information. • Analyze how context affects embedding choice.	ESE/LAB
LO 5.1: *Recall the architecture of neural models like RNN, LSTM, and GRU. (P.I.- 1.4.1 and 12.1.1) (CO1)*	• Label the architecture of RNN, LSTM, GRU. • Define gates and activation in each model. • Identify differences between models.	ISA/MSE/ESE
LO 5.2: *Differentiate between greedy decoding, beam search, and sampling strategies. (P.I.- 2.1.2 and 4.3.1) (CO2)*	• Define decoding strategies. • Compare them using examples or charts. • Select appropriate strategy for a task.	MSE/ESE
LO 5.3: *Apply neural models to generate or predict sequences in NLP tasks. (P.I.- 3.4.2 and 5.1.1) (CO3)*	• Use RNNs to generate sentences. • Train a model on sequence data. • Analyze generated output.	ESE/LAB
LO 5.4: *Evaluate language models based on accuracy, fluency, and relevance in output. (P.I.- 4.3.2 and 3.3.1) (CO5)*	• Define metrics like perplexity and BLEU. • Analyze fluency and relevance of predictions. • Compare performance of different language models.	ESE
LO 6.1: *Describe common NLP applications such as MT, IR, and sentiment analysis. (P.I.- 1.4.1 and 6.1.1) (CO1)*	• Define applications like MT, IR, QA. • Match applications to models and methods. • Give examples of use in regional languages.	ISA/MSE/ESE
LO 6.2: *Explain the workflow of an NLP application using deep learning models. (P.I.- 3.1.1 and 2.2.1) (CO2)*	• Outline each step of the pipeline. • Relate model type to task (e.g., LSTM for sentiment).	MSE/ESE/LAB

	• Interpret output at each pipeline stage.	
LO 6.3: *Assess the performance of your NLP system using relevant metrics. (P.I.- 3.3.1 and 4.3.2) (CO5)*	• Evaluate systems using F1, accuracy, BLEU. • Interpret output in light of those metrics. • Suggest improvements based on evaluation.	ESE/LAB
LO 6.4: *Develop an end-to-end NLP solution for a real-world problem. (P.I.- 3.2.2 and 5.2.1) (CO6)*	• Design and implement a complete NLP pipeline. • Use appropriate datasets and models. • Present and evaluate the final output.	ESE/LAB